

27-2 Port Security Troubleshooting Lab

Exercise

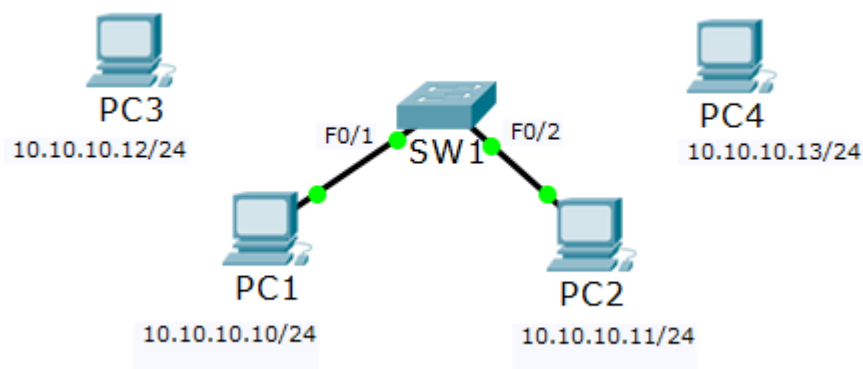
In this lab you will troubleshoot Port Security on a small campus network.

This lab continues on from the previous '28-1 Port Security Configuration'. Please ensure you have configured the Port Security settings in that lab.

Two additional PCs have been added, PC3 and PC4. The new PCs are not currently connected to the network.

In order for the lab to work as described please follow the steps in the exact order shown.

Lab Topology



Load the Startup Configurations

Open the '27-2 Port Security Troubleshooting.pkt' file in Packet Tracer to load the lab.

Port Security Troubleshooting

- 1) Verify that you can ping PC1 from PC2.
- 2) In Packet Tracer, delete the Ethernet cable connecting SW1 interface FastEthernet 0/2 to PC2.
- 3) Connect interface FastEthernet 0/2 on SW1 to the FastEthernet0 port on PC4.
- 4) Do you expect a ping from the PC4 to PC1 at 10.10.10.10 to succeed? Why or why not? Verify this.
- 5) In Packet Tracer, delete the Ethernet cable connecting SW1 interface FastEthernet 0/2 to PC4.
- 6) Enable Port Security sticky MAC address learning on SW1 interface FastEthernet 0/2.
- 7) Connect interface FastEthernet 0/2 on SW1 to the FastEthernet0 port on PC2.
- 8) Do you expect a ping from the PC2 to PC1 at 10.10.10.10 to succeed? Why or why not? Verify this.
- 9) In Packet Tracer, delete the Ethernet cable connecting SW1 interface FastEthernet 0/2 to PC2.
- 10) Connect interface FastEthernet 0/2 on SW1 to the FastEthernet0 port on PC4.
- 11) Do you expect a ping from the PC4 to PC1 at 10.10.10.10 to succeed? Why or why not? Verify this.
- 12) What is the Port Security status on SW1 interface FastEthernet 0/2?
- 13) In Packet Tracer, delete the Ethernet cable connecting SW1 interface FastEthernet 0/2 to PC4.
- 14) Connect interface FastEthernet 0/2 on SW1 to the FastEthernet0 port on PC2.
- 15) Do you expect a ping from the PC2 to PC1 at 10.10.10.10 to succeed? Why or why not? Verify this.

- 16) Ensure that PC2 can ping PC1. Do not change any 'switchport port-security' commands to do this.
- 17) **PC1's MAC address 0000.1111.1111 has been manually added to the Port Security configuration on SW1 interface FastEthernet 0/1.**
- In Packet Tracer, delete the Ethernet cable connecting SW1 interface FastEthernet 0/1 to PC1.
- 18) Connect interface FastEthernet 0/1 on SW1 to the FastEthernet0 port on PC3.
- 19) Do you expect a ping from the PC3 to PC2 at 10.10.10.11 to succeed? Why or why not? Verify this.